

# Day 4: Firewall Basics - Answers

Topic: Firewall types and use cases (Windows Defender Firewall)

## Q1. Spotify desktop app ka saara traffic block karna (Outbound Rule)

1. Start menu mein **Windows Defender Firewall with Advanced Security** search karke open karo.
2. Left panel mein **Outbound Rules** par click karo, phir right panel mein **New Rule...** select karo.
3. Rule Type: **Program** chuno, **Next**.
4. **This program path** choose karke Browse karo aur Spotify.exe select karo (usually C:\Users\<user>\AppData\Roaming\Spotify\Spotify.exe). **Next**.
5. Action: **Block the connection** chuno. **Next**.
6. Profile: **Domain, Private, Public** teeno tick rakho. **Next**.
7. Name do, jaise *Block Spotify Outbound*, aur **Finish** karo.
8. Test: Spotify kholo - songs load nahi hone chahiye / 'no internet' error aana chahiye.

*Note: Agar Spotify Microsoft Store version hai to uska path alag hota hai (WindowsApps folder).*

## Q2. Google Chrome ke liye Inbound Rule - sirf port 443 (HTTPS)

1. **Windows Defender Firewall with Advanced Security** kholo.
2. **Inbound Rules > New Rule...** click karo.
3. Rule Type: **Program > Next**.
4. Program path: C:\Program Files\Google\Chrome\Application\chrome.exe select karo. **Next**.
5. Action: **Allow the connection. Next**.
6. Profile select karo (Domain/Private/Public). **Next**. Name do: *Chrome Inbound 443*, **Finish**.
7. Ab banaye hue rule par **double-click** karo > **Protocols and Ports** tab kholo.
8. Protocol type: **TCP**, Local port: **Specific Ports > 443** likho. **Apply > OK**.

*Note: Windows ka default inbound behavior already 'block' hota hai, isliye sirf yeh allow rule hone se Chrome ke liye baaki ports par incoming connections block rahenge. Normal browsing mein Chrome outbound 443 use karta hai, isliye inbound rule practically kam hi zaroori hota hai.*

## Q3. Specific IP (123.45.67.89) ka saara traffic block karna

1. **Windows Defender Firewall with Advanced Security** open karo.
2. **Inbound Rules > New Rule...** (us IP se aane wala traffic block karne ke liye).
3. Rule Type: **Custom > Next**.
4. Program: **All programs > Next**.
5. Protocol and Ports: Protocol type **Any** rakho > **Next**.
6. **Scope** screen: 'Which remote IP addresses does this rule apply to?' mein **These IP addresses** select karo > **Add...** > *This IP address* mein **123.45.67.89** likho > **OK > Next**.
7. Action: **Block the connection > Next**.
8. Profile: teeno (Domain, Private, Public) > **Next**.

9. Name: *Block IP 123.45.67.89* > **Finish**.

10. **Outbound Rules** mein bhi same steps repeat karo, taaki aapka computer bhi us IP se connect na kar sake.

11. Verify: ping 123.45.67.89 chalao - reply nahi aana chahiye. Rule list mein green block icon dikhega.

## Q4. Hardware Firewall vs Software Firewall

| Point                   | Hardware Firewall                                                                                                      | Software Firewall                                                                                                         |
|-------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 1. Kahan lagta hai      | Separate physical device jo network aur internet ke beech (gateway par) lagta hai. Poore network ko protect karta hai. | Individual computer/server par installed program. Sirf usi device ko protect karta hai.                                   |
| 2. Performance & cost   | High throughput, dedicated processor; mehnga (device + license + maintenance).                                         | Device ke CPU/RAM use karta hai, thoda slow kar sakta hai; sasta ya free (OS ke saath built-in).                          |
| 3. Control & management | Centralized management, ek jagah se saare users ke rules; advanced features (VPN, IPS, content filtering).             | Har device par alag configure karna padta hai; per-application rules aasani se bante hain.                                |
| Use case                | Badi office ya data center jahan 100+ computers ko ek saath protect karna ho aur company-wide policy lagani ho.        | Personal laptop ya work-from-home / cafe Wi-Fi par - jahan har app ke liye alag rule chahiye (jaise Spotify block karna). |